How to write a report
session 2

Bioanalysis (FA-BA319/FA-CPS337)

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Based on: Linda McPhee, Linda Silvertand

Pictures from www.unsplash.com

Content:

Session 1 (May 27th):

1

- General
- Structure
- **Draft (*poster*)**
- Introduction: *context and importance*

Session 2 (June 10th):

2

- **Introduction**
- Material & Methods
- **Results** & Discussion
- Conclusion
- Abstract

Outline
Introduction (literature review)
Describe and visualize at least 1 result

The process

Content

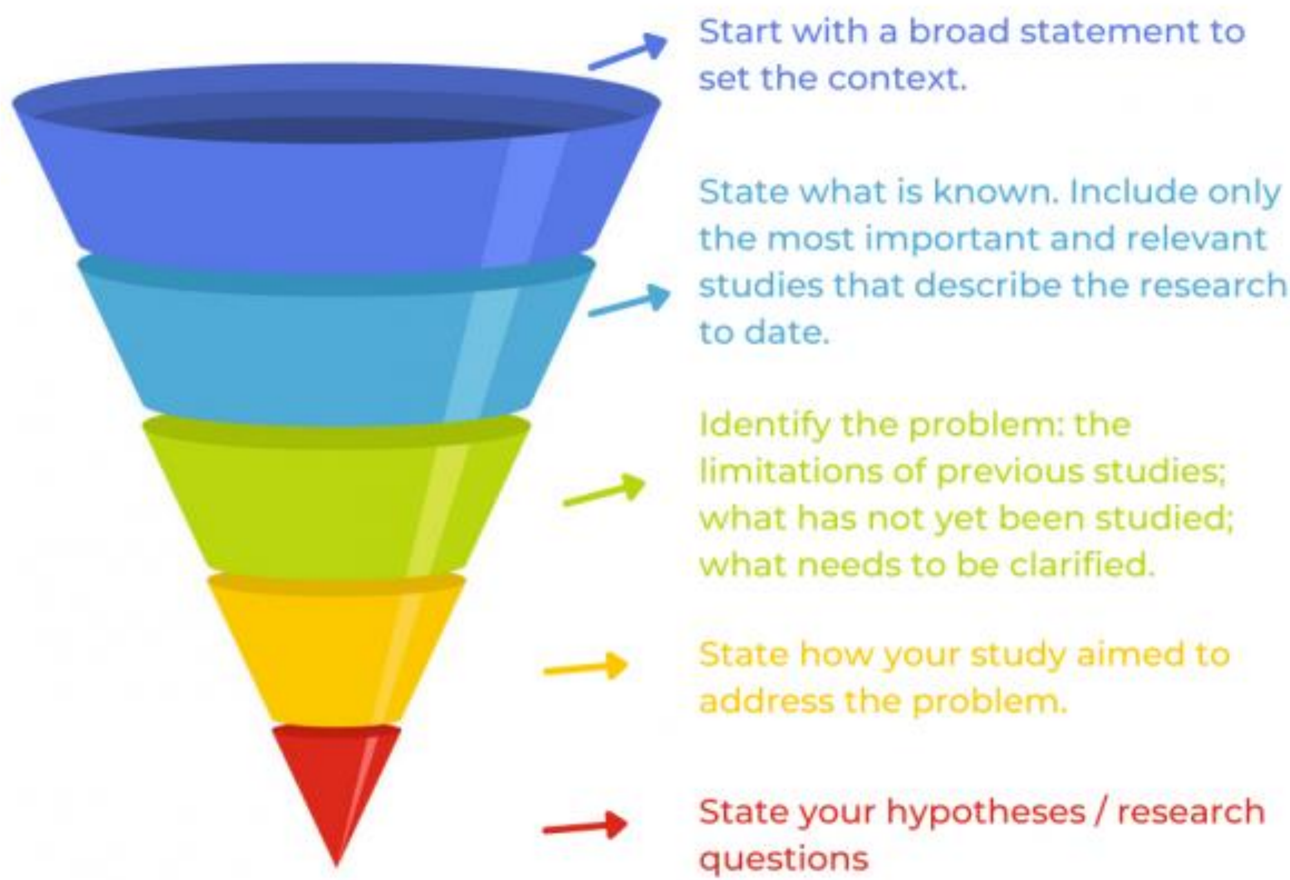
Outline

Draft

Report

Coherent story

Introduction: content and structure



https://asiaedit.com/blog/3-fundamental-principles-of-writing-an-effective-introduction-to-your-journal-article

Introduction: 5 questions

Bigger picture:

1. Set the context. Why is your research important?

Literature survey:

2. What is known?
3. What is unknown?

What has already been done?

What needs to be improved?

Your research:

4. How does your research fill these discrepancies?
5. What are your most important findings? Why should the reader read your paper?

➢ *Encourage the reader to read on!*

Introduction: 5 questions

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1. X can be used as a biomarker for…
2. Different analytical techniques (e.g.,….) have been developed to quantify X in…[]. Extraction of X from … often includes …[].
3. However, … methods showed poor sensitivity and recovery [], whereas commonly applied extraction methods are laborious and time-consuming [].
4. Technique Y enables … and can allow…
5. Using technique Y, we demonstrated … when applied to ….

The introduction is a STORY…

A story:

- Is **coherent**; the topics are all related, the use of language and layout is the same throughout the introduction, etc
- Has a certain **flow**; the order of the topics is logical and directed towards the revelation of your data and conclusions. The topics flow into each other smoothly.
- **Readable**: the language should not be a barrier in understanding of the content. Avoid complex use of language and long sentences. Check for grammar, style, etc (read, read, read and then read again).
- **Clear**: leave nothing open for the interpretation of the reader. Not everyone can read between the lines or interprets subjects the same way you do.
- **Is adjusted to the reader**. Know your reader.

Get your 5 answers and mold this into a story

Bigger picture:

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Sales pitch

Create your unique selling point

<u>Group:</u>

- Take your literature review and discuss:
  - What is unknown? What are current limitations?
  - How does your research fill this discrepancy?
  - What *details* do you want to share to get the reader curious and encourage the reader to read on?

➢ **Write your sales pitch (5 sentences):**

If you don't have any convincing results (yet), try to formulate your results/details in a positive way!
e.g., This method is a first step towards…
Despite challenges… further optimization will enable…

➢ Post your *sales pitch* in the Teams channel ☺

Introduction – final remarks

- Writing the introduction is easy, **editing** takes the most time and effort since there is a lot of reading and evaluating involved.
- Start at least **3-4 weeks (now!)** before handing in the paper with the introduction.
- It saves you unnecessary experiments, gets you a clear focus, but mainly saves you the trouble of a required second version.
- **DO NOT** write the paper in one go, briefly check for any major inconsistencies and then hand it in.

Materials and Method

What have you done to acquire the data?

- This couples the **reason** for the paper (introduction) to the **proof** (results)
- This is where you make your story believable
  - Formulate systematically all the experiments that were performed
  - Make it possible for someone else to repeat these experiments and get the same results

- **Not** a lab book!!! So:
  - No protocols (do not use the imperative)
  - Be concise
  - Use references (but do not refer to your lab book!)
  - Describe the changes you made to methods from literature

Materials and Method

- Consider: What is **THE** method?
- Use subsections:
  - Materials (describe quality and manufacturer)
  - Instruments (Make and type, sometimes also parts)
  - **THE** method: conditions, settings
    - Sample preparation
    - Separation
    - Detection
    - Quantification / calibration / internal standardization
  - Validation
  - Application
- Keep it factual (yes, that means it's a bit *boring*…)
- Check **examples**!

Examples

2 Materials and methods

2.1 Materials

Pharmalyte™ (36% w/v, pH 3–10, copolymer of glycine, glycylglycine, various amines and epichlorohydrin) was purchased from Amersham Bioscience (Piscataway, NJ, USA). Tween 20 (for synthesis) was obtained from Merck KGaA (Darmstadt, Germany). Hydroxyethyl cellulose (HEC, medium viscosity), phosphoric acid (85%), TFA (>99%), sinapic acid (>99%), TEMED (~99%), sodium hydroxide and all proteins (see Table 1) were purchased from Sigma-Aldrich. GlucaGen (Novo Nordisk) was kindly donated by Drs L. J. F. Silvertand and Drs F. N. Sikke (Apotheek Born, Born, The Netherlands). HPLC gradient grade ACN and methanol (MeOH) were purchased from Biosolve (Valkenswaard, The Netherlands). High-purity water (Ultrapure, >18.2 MΩcm) was obtained from a Millipore Synergy UV water purification system (Millipore, Bedford, MA, USA).

2.2 Apparatus and methods

cIEF separations were performed with a PrinCE 500 dual-lift CE system using Dax Data Acquisition and Analysis Software version 6.0 (Prince Technologies, Emmen, The Netherlands), equipped with a Knauer K-2501 UV detector set at 280 nm (Knauer GmbH, Berlin, Germany) and an eCAP neutral capillary (50 μm id, 40.2 cm total length, Beckman Coulter, Fullerton, CA, USA). Prior to each run the capillary was rinsed first with 10 mM phosphoric acid in 0.3% HEC, then with purified water and finally with 0.1 % Tween 20 in 0.3% HEC. Subsequently, the capillary was filled with Sample 1 (ribonuclease (RNase, 90 μg/mL), myoglobin (Myo), carbonic anhydrase II (CAII), β-lactoglobulin B (LGB) and α-lactalbumin (Lac) (all at 10 μg/mL), Pharmalyte™ (1% w/v), TEMED (0.1% v/v), HEC (0.3% w/v) and Tween 20 (0.1% w/v)) or with Sample 2 (= Sample 1, but LGB and Lac are replaced with β-lactoglobulin A (LGA) (16 μg/mL) and cholecystokinin flanking peptide (CCK) (8 μg/mL)). The inlet vial (anolyte) was filled with 10 mM phosphoric acid in 0.3% HEC. The capillary outlet was inserted in the hollow needle (electrically grounded) of the CE interface of an LC Packings Probot Microfraction Collector (Dionex, Amsterdam, The Netherlands) (see also Fig. 1). Sheath liquid (200 mM ammonium hydroxide in 50% MeOH and 0.01% Tween 20) was delivered through this needle *via* the syringe of the Probot Spotter and was set to deposit 0.5 μL sheath liquid *per* spot throughout the entire run. During focusing the capillary outlet and needle were submerged in the outlet vial (catholyte: 200 mM ammonium hydroxide in 50% MeOH). Focusing was performed at 20 kV at room temperature for 15–90 min. After focusing, the catholyte waste vial was removed and the separated proteins were mobilized at 20 V and 100 mbar air pressure towards an ABI stainless steel MALDI target plate (Applied Biosystems, Framingham, MA, USA) with 192+6 predefined spots. Upon completion of the spotting procedure, 0.75 μL MALDI matrix (5 mg/mL sinapic acid in 80% v/v ACN with 0.1% v/v TFA) was added *per* spot and dried.

MALDI-TOF MS experiments were performed with an Applied Biosystems 4700 Proteomics Analyzer. The m/z recording range was set to 5000–40 000 with a focus m/z of 20 000 in linear mode. MALDI-TOF spectra were analyzed using Data Explorer software Version 3.0 (Applied Biosystems).

Examples

Instrumentation

The LC system consisted of a model 1100 series binary LC pump, an 1100 series cooled autosampler (10°C) and an 1100 series mobile phase degasser (all Agilent Technologies, Palo Alto, CA, USA). A stepwise gradient was applied to elute the analytes. At time zero a mixture of 96% 1 mM ammonia solution and 4% acetonitrile was flushed through the column. After 2 min, the percentage of acetonitrile was increased to 25% in 1 min and maintained at this percentage for 3 min. From 6 to 6.1 min, the percentage of acetonitrile was set back to 4% and was pumped through the column during the last 6 min of the run. The flow was set at 400 µL/min. The analytes were separated on an Extend C18 column (150 × 2.1 mm i.d., 5 µm particle size), with an Extend C18 pre-column (12.5 × 2.1 mm i.d., 5 µm particle size) (Agilent Technologies) protected by a 5 µm in-line filter (Upchurch Scientific, Oak Harbor, WA, USA). The column was connected to the ESI source of a API 3000 triple quadrupole mass spectrometer (PE Sciex, Thornhill, ON, Canada) through a divert valve (Valco Instruments, Houston, TX, USA). The valve was used to prevent the introduction of endogenous compounds into the mass spectrometer by directing the eluent flow the first 2 min and the last minute to waste. The injection volume was 10 µL. Multiple reaction monitoring (MRM) chromatograms were used for quantification using Analyst™ software version 1.2 (PE Sciex). Standard solutions of the compounds were infused directly into the ESI source to determine positively charged precursor ions. Only singly charged ions were observed. Product ions were determined by subjecting the precursor ions to collision-induced dissociation. Transitions of the precursor-product ion pairs are listed in Table 1 and ESI-MS/MS settings are listed in Table 2.

Table 2. MS/MS settings

Parameter	Setting
Duration	12 min
Ionspray voltage (positive ion mode)	+5 kV
Curtain gas (N_2)	1.2 L/min
Temperature	400°C
Nebulizer gas (compressed air)	1.8 L/min
Turbo gas (N_2)	7.0 L/min
Collision-activated dissociation gas (N_2)	8 psi
Channel electron multiplier	1900 V

Parameter	Fludarabine	Cladribine	Cyclophosphamide	d4-Cyclophosphamide
Dwell time (ms)	50	50	50	50
Declustering potential (V)	36	31	41	56
Focusing potential (V)	160	150	180	240
Entrance potential (V)	10	10	10	10
Collision energy (V)	25	21	31	31
Collision cell exit potential (V)	10	12	36	26

Rapid Commun. Mass Spectrom. 2005; **19**: 3673–3680

Examples

Chemicals

Cyclophosphamide monohydrate reference standard and d4-cyclophosphamide (Fig. 1(D)), internal standard for cyclophosphamide, originated from Baxter Oncology GmbH (Frankfurt am Main, Germany). Fludarabine reference standard was obtained from Sigma-Aldrich Chemie GmbH (Steinhein, Germany). The Leustatin® formulation (cladribine, Fig. 1(B)) was used as the internal standard for fludarabine and was purchased from Janssen-Cilag BV (Tilburg, The Netherlands). Acetonitrile and methanol (both supra-gradient grade) were from Biosolve Ltd. (Amsterdam, The Netherlands). Ammonium acetate, 25% ammonia, 100% acetic acid, 37% hydrochloric acid (all analytical grade) and LiChrosolve water (LC gradient grade) originated from Merck (Darmstadt, Germany). Drug-free human heparin control plasma and drug-free human EDTA control plasma originated from Sanquin (Amsterdam, The Netherlands).

Preparation of stock solutions, working solutions, calibration standards and quality control samples

Stock solutions of cyclophosphamide were prepared independently by dissolving cyclophosphamide monohydrate in methanol at a concentration of 1.0 mg/mL. One solution was used to spike plasma calibration standards and one was prepared independently to spike quality control (QC) samples. The stock solutions for fludarabine were prepared in acidified methanol (10 mM HCl in methanol).

Working solutions were prepared from stock solutions and contained both cyclophosphamide and fludarabine at concentration levels of 300, 2500 and 7500 ng/mL in methanol for the preparation of QC samples and 100, 200, 500, 1000, 5000 and 10 000 ng/mL for the preparation of calibration standards. All solutions were stored at −20°C. Plasma QC samples and calibration standards were prepared by adding working solutions to control human heparin plasma or control human EDTA plasma. QC samples were prepared at low, medium and high concentration levels (3, 25 and 75 ng/mL) and additional levels around the lower limit of quantitation (LLQ, 1 ng/mL) and at a concentration equal to 25 times the upper limit of quantitation (ULQ, 100 ng/mL). Calibration working solutions were used to spike human control plasma to obtain concentration levels for calibration standards of 1, 2, 5, 10, 50, 75 and 100 ng/mL.

A stock solution of d4-cyclophosphamide was prepared by dissolving d4-cyclophosphamide in methanol at a concentration of 1 mg/mL. A working solution of d4-cyclophosphamide was prepared by diluting the stock solution with methanol to a concentration of 40 µg/mL. Cladribine was already in solution at a concentration of 1 mg/mL (Leustatin®). This stock solution was diluted with methanol to obtain a cladribine working solution of 20 µg/mL. The final working solution was obtained by adding 100 µL 40 µg/mL d4-cyclophosphamide working solution to 100 µL 20 µg/mL cladribine working solution and adding methanol up to 10.0 mL. Plasma standards and samples and stock solutions of fludarabine, cladribine, cyclophosphamide and d4-cyclophosphamide were stored at −20°C.

Results

Describe AND Present

- The results (and discussion) section is a **STORY** as well.
- The text should be used to **guide the reader**:
  - Use tables when **exact numbers** are important and are too much / complex to describe.
  - Use figures when…an **image is worth a thousand words**!
  - Refer to tables and figure in the text.
  - The text should **not** simply repeat the information in the tables and figures.
  - So, **not** just a series of tables without explanation!

- Be **selective**! **Never** *present* the (same) data in more than one way!

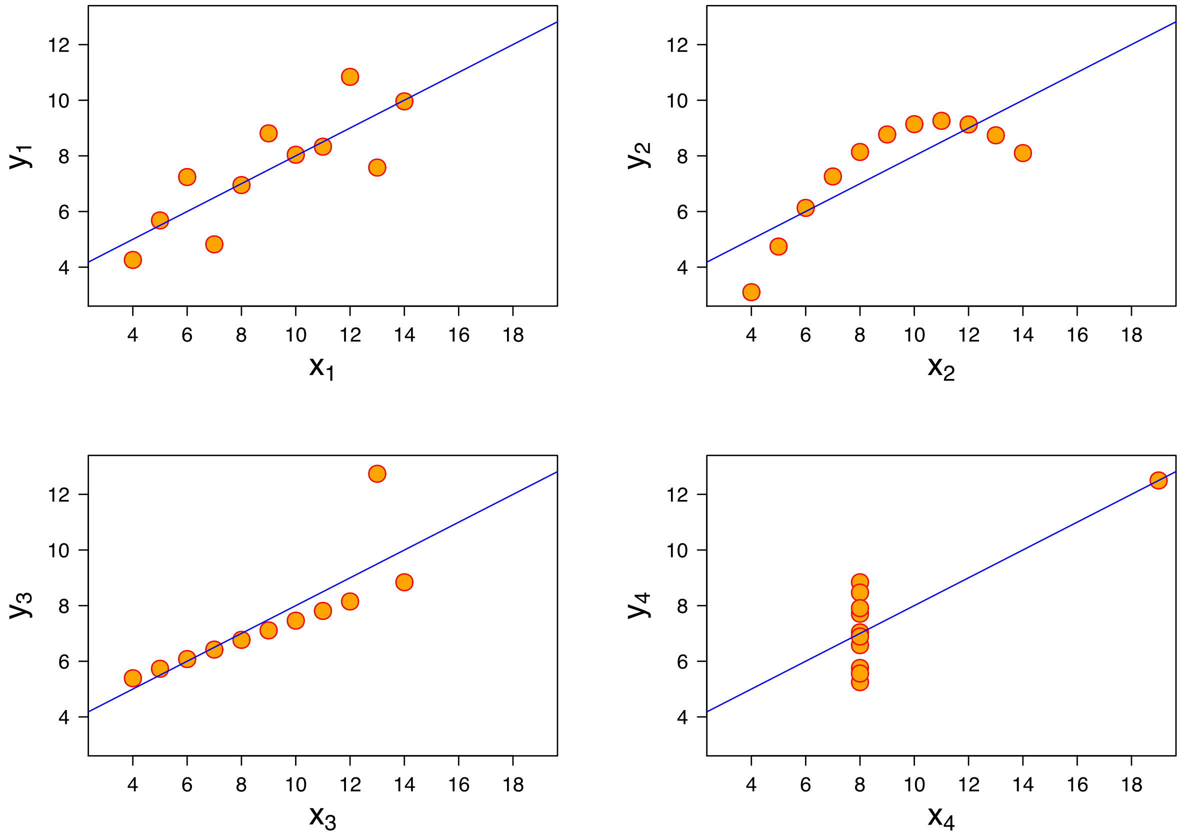
Visualizing Results

	I		II		III		IV	
	x	y	x	y	x	y	x	y
	10	8,04	10	9,14	10	7,46	8	6,58
	8	6,95	8	8,14	8	6,77	8	5,76
	13	7,58	13	8,74	13	12,74	8	7,71
	9	8,81	9	8,77	9	7,11	8	8,84
	11	8,33	11	9,26	11	7,81	8	8,47
	14	9,96	14	8,1	14	8,84	8	7,04
	6	7,24	6	6,13	6	6,08	8	5,25
	4	4,26	4	3,1	4	5,39	19	12,5
	12	10,84	12	9,13	12	8,15	8	5,56
	7	4,82	7	7,26	7	6,42	8	7,91
	5	5,68	5	4,74	5	5,73	8	6,89
SUM	99,00	82,51	99,00	82,51	99,00	82,50	99,00	82,51
AVG	9,00	7,50	9,00	7,50	9,00	7,50	9,00	7,50
STDEV	3,32	2,03	3,32	2,03	3,32	2,03	3,32	2,03

Anscombe FJ (1973). *Graphs in Statistical Analysis*. The American Statistician 27, 1, 17–21.

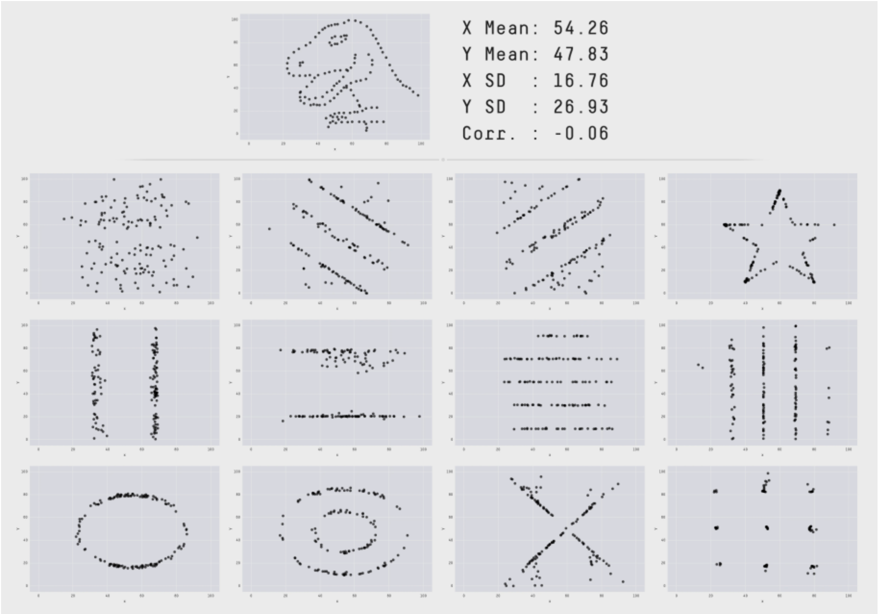
Visualizing Results

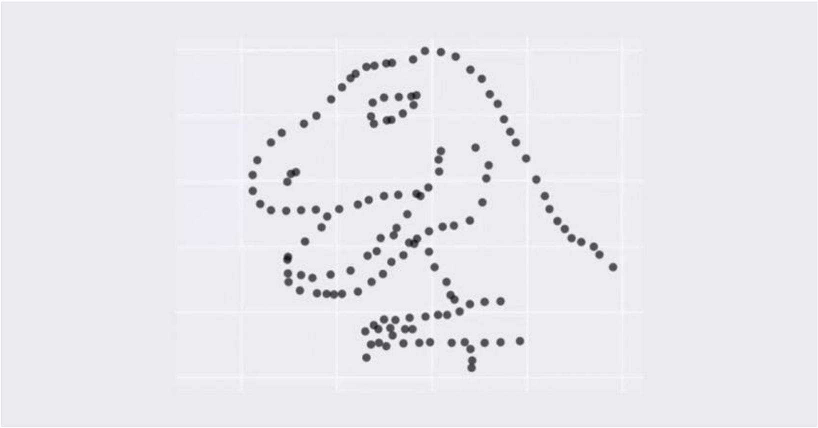
A picture is worth a thousand words



Anscombe FJ (1973). *Graphs in Statistical Analysis*. The American Statistician 27, 1, 17–21.

The Datasaurus Dozen ☺:
Never trust summary statistics alone!





Matejka J & Fitzmaurice G (2017). *Same Stats, Different Graphs: Generating Datasets with Varied Appearance and Identical Statistics through Simulated Annealing*. https://www.autodeskresearch.com/publications/samestats.

Results: Tables

Use tables when **exact numbers** are important and are too much / complex to describe.

- Provide the table with a **descriptive title**
- Avoid redundancy

➢ the reader should be able to understand the table without reading the text!

Table 4: Average extraction recovery, ion suppression and total recovery (n=3) for Enfuvirtide, Tifuvirtide and M-20 at 3 concentration levels

Nominal concentration (ng/ml)	Extraction recovery (%)	Ion suppression (%)	Total recovery (%)
Enfuvirtide			
51.27	88.5 ± 9.9	−7.2 ± 0.9	82.2 ± 4.9
2000	89.2 ± 9.7	−10.8 ± 1.3	79.6 ± 6.9
7691	87.1 ± 3.1	−15.6 ± 0.6	73.5 ± 2.7
Tifuvirtide			
51.26	94.3 ± 16.1	−11.4 ± 2.1	83.6 ± 13.8
1999	82.8 ± 8.9	−8.5 ± 1.1	75.8 ± 8.0
7690	81.4 ± 3.1	−16.6 ± 0.6	67.8 ± 2.4
M-20			
19.40	91.5 ± 11.2	−3.2 ± 0.5	88.6 ± 9.8
378	96.6 ± 11.6	−7.9 ± 1.2	88.9 ± 12.4
1455	103.0 ± 5.0	−16.7 ± 1.0	85.8 ± 4.3

Results: Figures

CE-UV

The first measurement (phenylalanine standard) gave the following results:

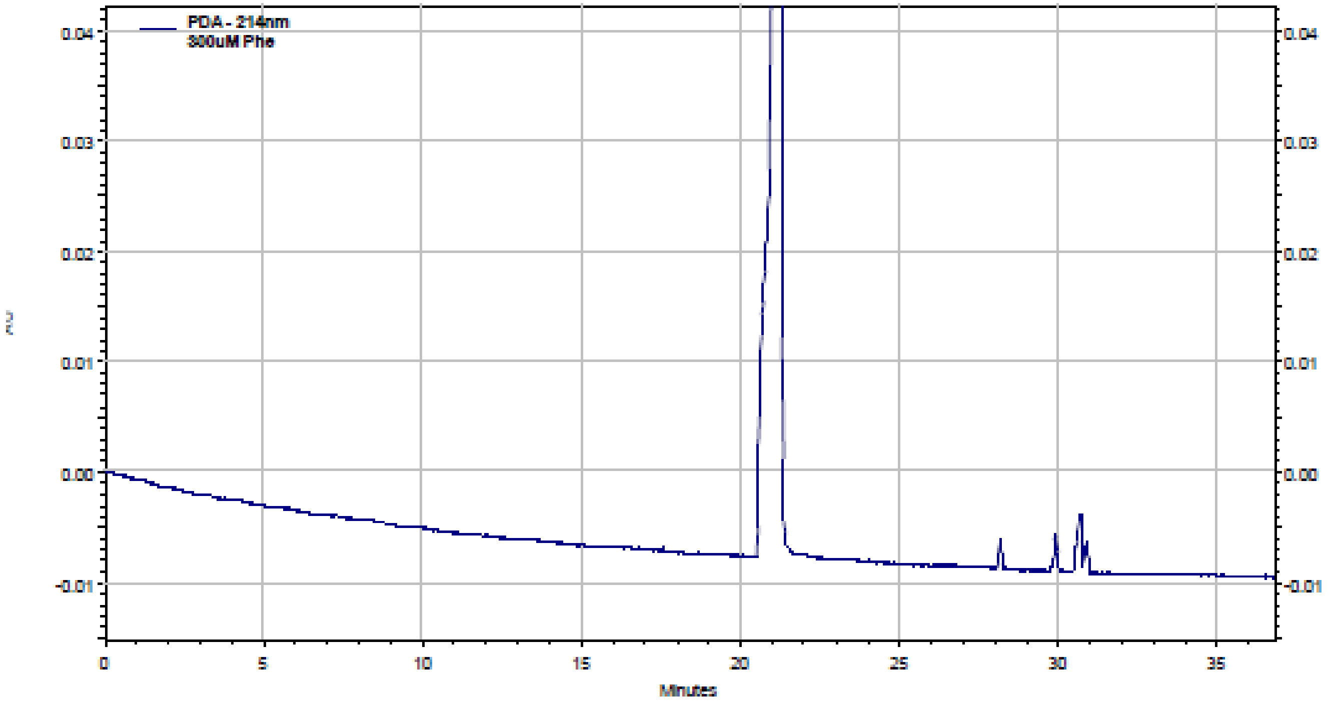


Figure 1: Electropherogram at 214 nm.

Results: Figures

CE-UV

CE-UV analysis of a … ng/mL phenylalanine standard in … showed a main peak with a retention time of … minutes (**Figure 1**).

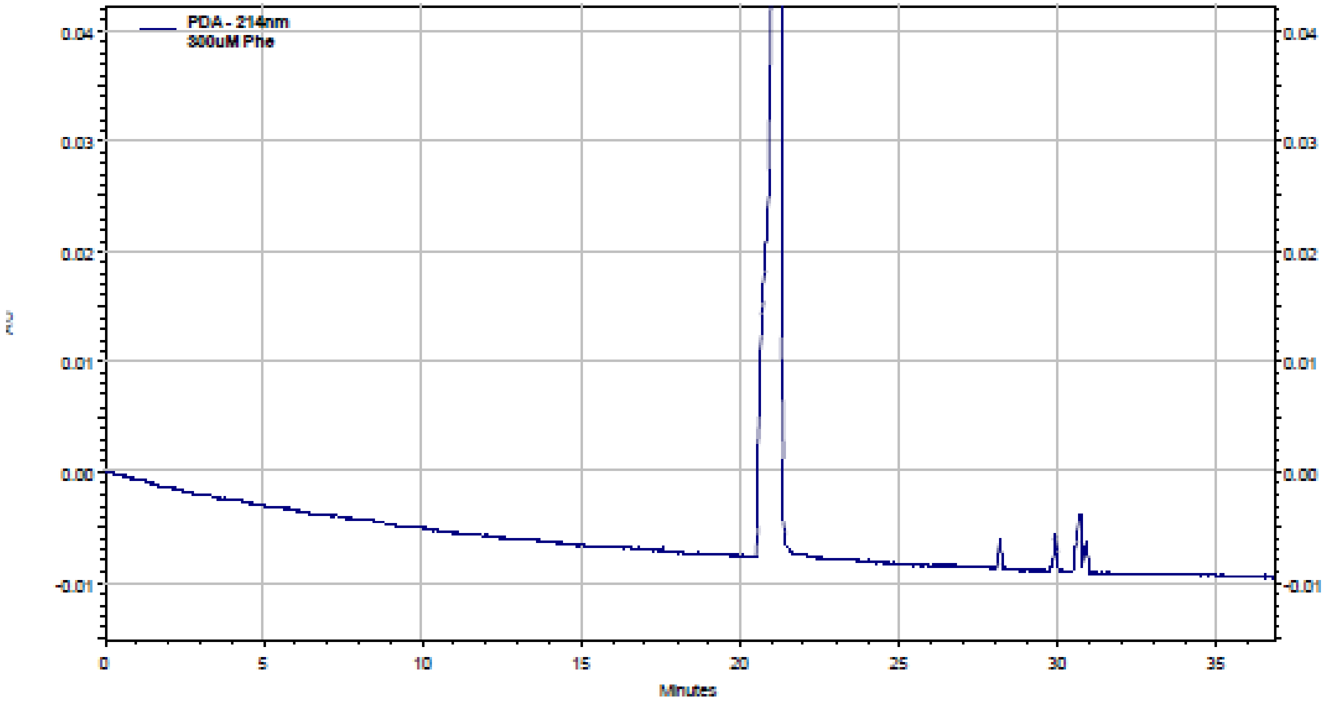


Figure 1: CE-UV electropherogram of phenylalanine (…ng/mL) at 214 nm.

Results: Description

The retention time (RT) of VPA with this method was determined to be 13.74 minutes (see figure 3B). In this chromatogram, a large peak was seen at around 13.74 minutes in the spiked sample that was not visible in the blank (see figure 3A). At 13.70 min there was a small peak visible in the blank sample, but this was not seen again after the further optimization of the GC protocol, and therefore did not interfere with the VPA signal. The tailing factor increased substantially for the DB-624UI column (see figure 4). Therefore, this column was used in all following experiments.

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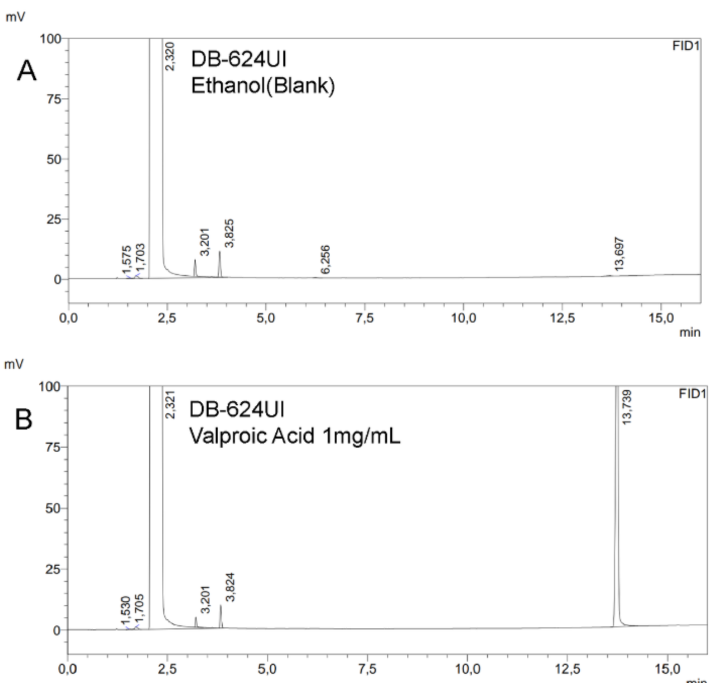
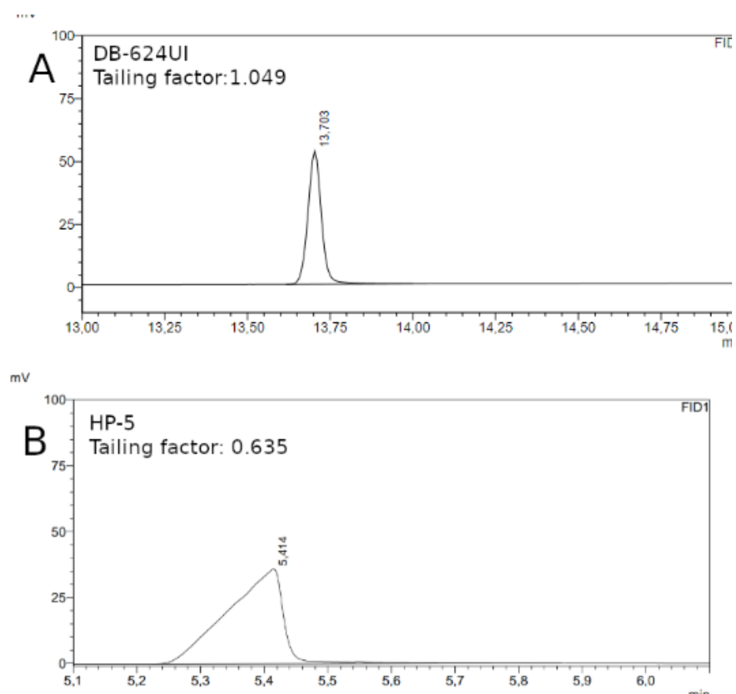




The retention time (RT) of VPA with **this method** was determined to be 13.74 minutes (see figure 3B). In **this chromatogram**, a **large** peak was seen at **around** 13.74 minutes in **the spiked sample** that was not visible in **the blank** (see figure 3A). At 13.70 min there was a **small** peak visible in **the blank** sample, but **this** was not seen again after further optimization of the GC protocol, and therefore did not interfere with the VPA signal. The tailing factor increased **substantially** for the DB-624UI column (see figure 4). Therefore, **this column** was used in **all following experiments**.

Analysis of *a standard with 1 mg/mL VPA in ethanol* showed that the retention time of VPA was 13.74 minutes (Figure 3a). Analysis of *blank ethanol* showed a peak at 13.70 minutes with *an AUC that was xxx%* of the AUC of the 1 mg/mL VPA standard. *[if it was not seen again, may be no need to mention it? Or explain what has been optimized?]*
The use of a *HP-5 column* resulted in an asymmetrical (*fronting*) peak *with a tailing factor of 0.635* (Figure 4). Changing to a DB-624UI column resulted in a symmetrical peak *with a tailing factor of 1.049* (Figure 4). Therefore, the DB-624UI column was selected.

Results – Exercise (1)

How clear is your description?

<u>Individually:</u>

- Describe and visualize at least 1 result

<u>In pairs (different groups!):</u>

- Exchange descriptions (*without* table/figure) and try to **recreate the result** from this description.
- **Evaluate:** is all the relevant information depicted in the reconstructed image? What information is missing in the description?

Results – Exercise (2)

How clear is your table/figure?

Individually:

- Select a table/figure based on your results
- Write your main conclusion down

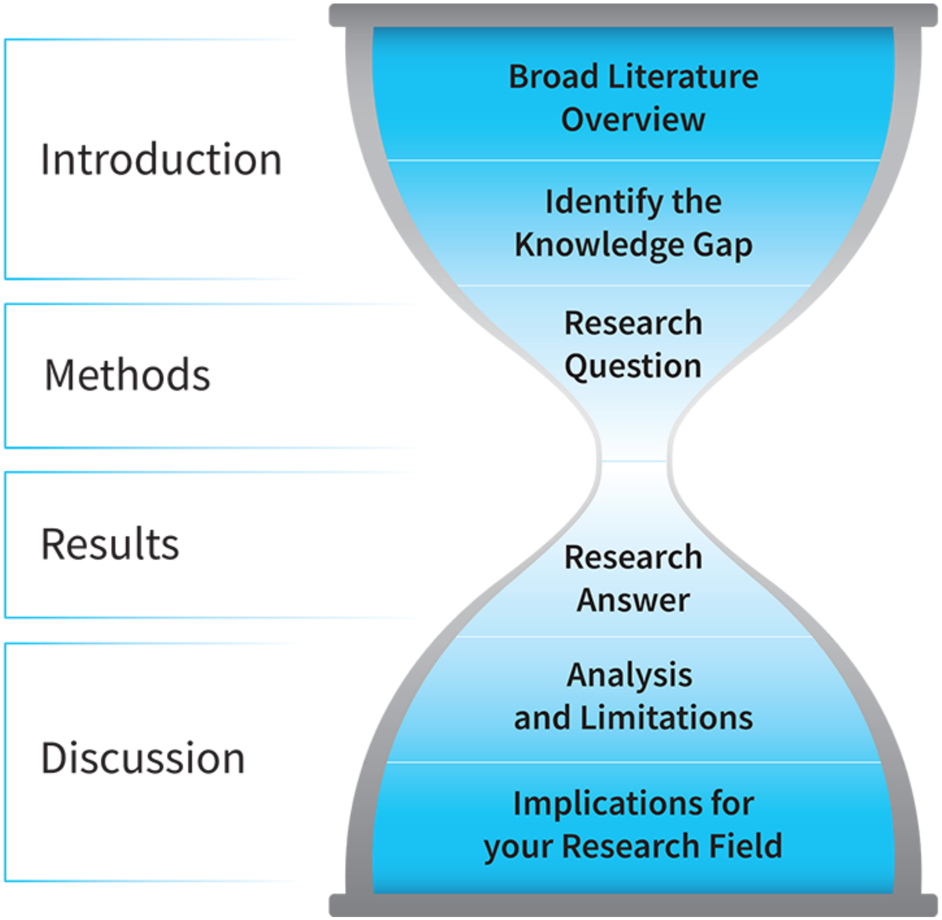
In pairs (different groups – different person!):

- Exchange your table/figure (without description – caption is ok!)
- Write down your conclusion based on the table/figure from someone else.
- **Evaluate:** Discuss the differences in interpretation. What information is needed to correctly interpret the result? (NB: this is the minimal information needed for your caption!)

Discussion

- The most important part of the report
  - So, no rushing this…
- **Together with the Results**:
  - More coherent
  - Easier for you and for the reader (everything together)
  - Use paragraphs / subsections!
  - Avoid that discussion is written as a summary of the results
- Instead, consider:
  - Do the results **match your expectations**?
  - Try to **explain unexpected results**
  - Place individual results in a wider context (combine results)
  - Discuss possibilities for improvement and suggest possible follow-up experiments

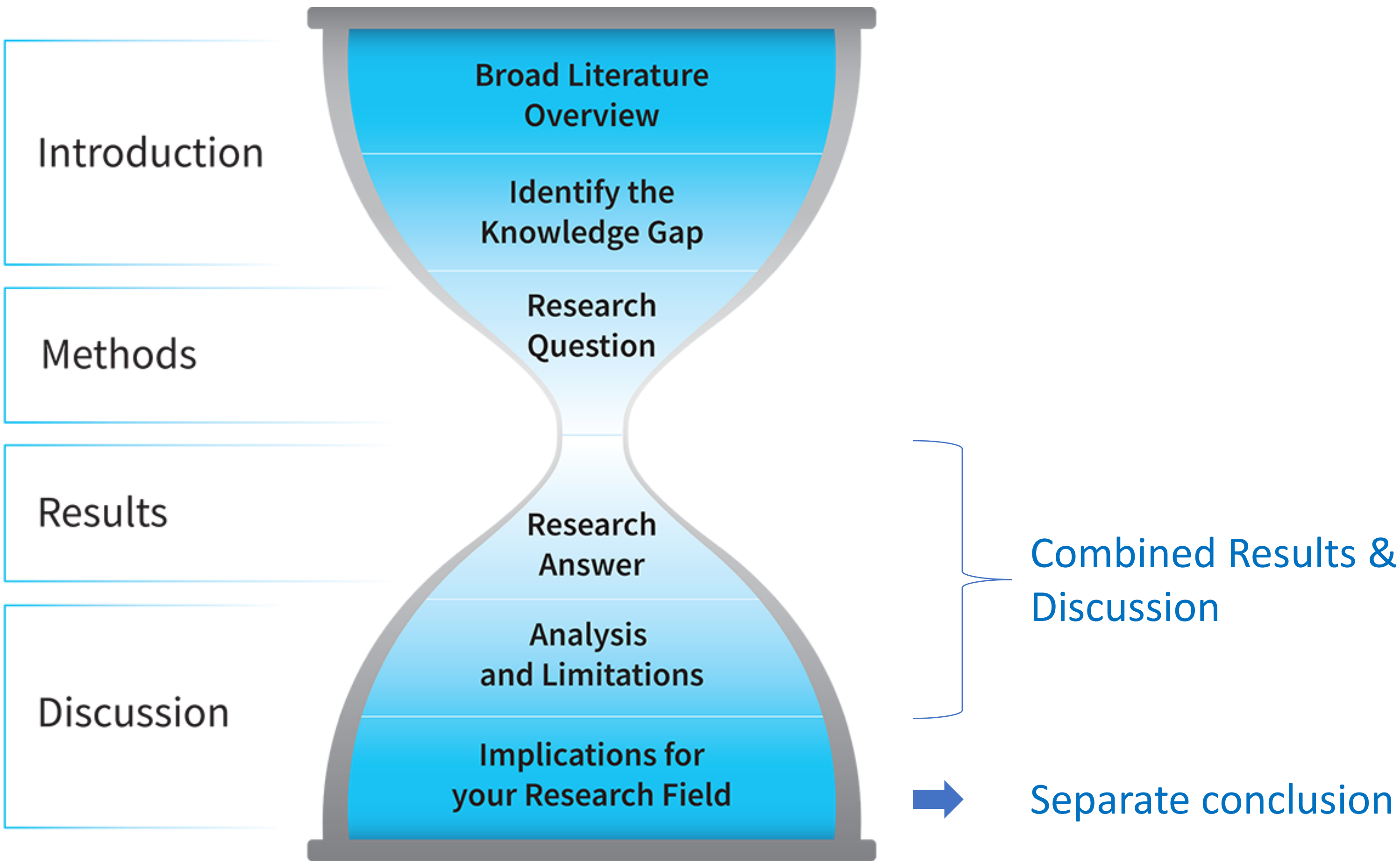
Discussion: Structure



Opposite to the Introduction!

https://www.adinstruments.com/blog/tips-writing-scientific-paper

Discussion: Structure



https://www.adinstruments.com/blog/tips-writing-scientific-paper

Results & Discussion

- What did you find?
- Is it surprising?
- What are the **similarities**, **differences** and **inconsistencies**:
  - between the different experiments? **Your opinion**
  - compared to groups that performed the same experiments? **Peer opinion**
  - compared to peer-reviewed literature on the same topic? **Expert opinion**
- Explain the **similarities**, **differences** and **inconsistencies**:

➢ Focus on the reliability of the data:

  - Are the methods reliable?
  - What data are most reliable
  - What are the limitations of your work?

Conclusion

- A conclusion is **NOT** a summary of your results and discussion, instead:

➢ **Generalise**
  - Assemble all data and formulate a weighted interpretation.
  - Mention also **the reliability** of these data
  - Try and find the most plausible explanation for your results, preferably underlined with peer-reviewed literature.

➢ **Consider the bigger picture**
  - What (global) implications does your research have?

➢ **Think about the next steps**
  - What would you recommend the next person to continue puruing the goal of your project?

Recommendations

- What will be the logical next steps in this research?
- How would you plan and perform these next steps?
- What are the shortcomings in your own research and how would you solve these?
- What data do you need to complete the entire story?
- What does the next person in the line of this project need tot do?

➢ To answer these questions, try to look ahead in the chain of the project

Summary / abstract

- **Attract the attention** of the reader in a haystack of other papers
- Be short, compact and **concise**
- Content (4-5 sentences):
  1. What have you been trying to do (or why did you want to do this) (goal)
  2. What DID you actually do (method)
  3. What did you find (results)
  4. What does this mean (discussion/conclusion)

Summary - example

GOAL

Fludarabine and cyclophosphamide are anticancer agents mainly used in the treatment of hematologic malignancies. We have developed and validated an assay using high-performance liquid chromatography (HPLC) coupled with electrospray ionization tandem mass spectrometry for the quantification of fludarabine in combination with cyclophosphamide in human heparin and human EDTA plasma. Sample pre-treatment consisted of a protein precipitation with cold acetonitrile (−20°C) using 250 μL of plasma. Separation was performed on an Extend C18 column (150 × 2.1 mm i.d.; 5 μm) with a stepwise gradient using 1 mM ammonia solution and acetonitrile at a flow rate of 400 μL/min. The analytical run time was 12 min. The triple quadrupole mass spectrometer was operated in the positive ion mode and multiple reaction monitoring was used for drug quantification. The method was validated over a concentration range of 1 to 100 ng/mL for fludarabine and cyclophosphamide in human heparin and human EDTA plasma. The coefficients of variation were <13.9% for inter- and intra-day precisions. Mean accuracies were also within the designated limits (±15%). The analytes were stable in plasma, processed extracts and in stock solution under all relevant conditions.

Summary - example

METHOD

Fludarabine and cyclophosphamide are anticancer agents mainly used in the treatment of hematologic malignancies. We have developed and validated an assay using high-performance liquid chromatography (HPLC) coupled with electrospray ionization tandem mass spectrometry for the quantification of fludarabine in combination with cyclophosphamide in human heparin and human EDTA plasma. Sample pre-treatment consisted of a protein precipitation with cold acetonitrile (−20°C) using 250 μL of plasma. Separation was performed on an Extend C18 column (150 × 2.1 mm i.d.; 5 μm) with a stepwise gradient using 1 mM ammonia solution and acetonitrile at a flow rate of 400 μL/min. The analytical run time was 12 min. The triple quadrupole mass spectrometer was operated in the positive ion mode and multiple reaction monitoring was used for drug quantification. The method was validated over a concentration range of 1 to 100 ng/mL for fludarabine and cyclophosphamide in human heparin and human EDTA plasma. The coefficients of variation were <13.9% for inter- and intra-day precisions. Mean accuracies were also within the designated limits (±15%). The analytes were stable in plasma, processed extracts and in stock solution under all relevant conditions.

Summary - example

RESULTS

Fludarabine and cyclophosphamide are anticancer agents mainly used in the treatment of hematologic malignancies. We have developed and validated an assay using high-performance liquid chromatography (HPLC) coupled with electrospray ionization tandem mass spectrometry for the quantification of fludarabine in combination with cyclophosphamide in human heparin and human EDTA plasma. Sample pre-treatment consisted of a protein precipitation with cold acetonitrile (−20°C) using 250 µL of plasma. Separation was performed on an Extend C18 column (150 × 2.1 mm i.d.; 5 µm) with a stepwise gradient using 1 mM ammonia solution and acetonitrile at a flow rate of 400 µL/min. The analytical run time was 12 min. The triple quadrupole mass spectrometer was operated in the positive ion mode and multiple reaction monitoring was used for drug quantification. The method was validated over a concentration range of 1 to 100 ng/mL for fludarabine and cyclophosphamide in human heparin and human EDTA plasma. The coefficients of variation were <13.9% for inter- and intra-day precisions. Mean accuracies were also within the designated limits (±15%). The analytes were stable in plasma, processed extracts and in stock solution under all relevant conditions.

Summary - example

CONCLUSION

Fludarabine and cyclophosphamide are anticancer agents mainly used in the treatment of hematologic malignancies. We have developed and validated an assay using high-performance liquid chromatography (HPLC) coupled with electrospray ionization tandem mass spectrometry for the quantification of fludarabine in combination with cyclophosphamide in human heparin and human EDTA plasma. Sample pre-treatment consisted of a protein precipitation with cold acetonitrile (-20°C) using 250 μL of plasma. Separation was performed on an Extend C18 column (150×2.1 mm i.d.; 5 μm) with a stepwise gradient using 1 mM ammonia solution and acetonitrile at a flow rate of 400 μL/min. The analytical run time was 12 min. The triple quadrupole mass spectrometer was operated in the positive ion mode and multiple reaction monitoring was used for drug quantification. The method was validated over a concentration range of 1 to 100 ng/mL for fludarabine and cyclophosphamide in human heparin and human EDTA plasma. The coefficients of variation were <13.9% for inter- and intra-day precisions. Mean accuracies were also within the designated limits (±15%). The analytes were stable in plasma, processed extracts and in stock solution under all relevant conditions.

References

- Use references as used in literature
- Number references and use numbers to refer
- Only list references you actually refer to in the text

See: https://libguides.library.uu.nl/citing

Appendices

- Only add appendices when relevant
- Only add appendices you refer to in the text
- Number appendices
- Appendices should also come with a title and legend

Feedback session 2

https://forms.gle/EdJt9F67WLyDWVGd9

- What did you expect from the report writing seminar(s)?
- To what extend was your expectation met in session 2?
- List at least 3 things from session 2 that were helpful for you
- What could be improved in the **structure** (exercises, interaction, time management) of session 2?
- What could be improved in the **content** (topics) of session 2?
- What topics did you miss in session 1 and 2?
- Any additional comments or feedback are welcome!

Session 1:

https://forms.gle/LvWjsToRhMYoYuS9A